ADA-PARC

Scorecard Information

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Theory & Process

Data Source

Where is the data coming from?

Data used for the development of ADA-PARC scorecards derives from the U.S. Census - American Community Survey (ACS) year 2023, five-year estimates.

Why Scorecards?

Purpose of ADA-PARC Scorecard development is to translate raw measures into easily interpretable and comparable indices of quality of life for the disabled population.Essentially what we are doing is evaluating, for each individual measure, how the disabled population in state X fares relative to the disabled population of the median state on that measure. An example of this would be the percentage of people with disabilities living below the poverty line in Illinois compared to the percentage of people with disabilities living below poverty in the median state on this metric.

Calculating Scores

So then how are raw values such as % living below the poverty line converted to a 100 point scale?

Step 1: Translate Raw Data to Z-Score

First, we calculate the z-score for each state on every measure. A **z-score** simply tells you how many standard deviations a value is above or below the mean of a dataset. It standardizes data so different values can be compared on the same scale.

Note: Z-scores are clipped between -3 and 3, and reversed if a higher score on an individual measure is undesirable (e.g. % below poverty). This allows us to keep a consistent scale such that for our final 100 point scores, 0 is the worst and 100 is the best.

Step 2: Translate Z-Score to 100-point Scale

In order to convert z-scores scaled from -3 to 3 into a 100 point scale we use a simple mathematical transformation. We multiply the proportion 50/3 by Z and add 50.

Z-Score → 100-point Score : 50/3z + 50

Step 3: Caclulate Index Variables using 100pt Scores

Many of our individual scorecard measures are then grouped together into indices which give insight into broader categories.

Example: Employment and Income

PWD % Employed Score + PWD Median Income Score / 2 = Employment and Income Score

Scores Calculation : Example

State Name	Raw Score	Z-Score	100pt Score
-	pwd_home_pct	z_pwd_home_pct	100pt_pwd_home
Pretendland	98%	2	**85**

Interpretation

For the purpose of interpretation of any tables and visual elements as part of the ADA-PARC scorecards, the following rules always apply:

- **Higher #: Better outcomes for people with disabilities**
- **Higher #: Represented by darker color**

Positive Indicator Example

State Name	Raw Score	Z-Score	100pt Score
-	pwd_median_income	z_pwd_median_income	Median income Score
Vermont	$23,809	-1.26	**33**
Hawaii	$37,779	2.09	**76**

In this condensed table, two states are displayed to highlight how scores are developed and can be compared.

- In Hawaii, the median income for the disabled population is $37,779 significantly better than average, thus corresponding to a z-score of +2.09 and a 100pt score of 76.

- In Vermont, the median income for the disabled population is $23,809 which is worse than average, thus corresponding to a z-score of -1.26 and a 100pt score of 33.

Negative Indicator Example

State Name	Raw Score	Z-Score	100pt Score
-	pwd_below_poverty_pct	z_pwd_below_poverty	100pt_pwd_below..
Utah	18.06	1.52..	**69**
Kentucky	30.83	-1.09..	**36**

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- 18% of the disabled population lives in Utah lives in poverty, since this is significantly better than average, Utah ends up with a positive z-score, and thus a 100pt score above 50.
- 30% of the disabled population in Kentucky lives in poverty, this is worse than average, corresponding to a z-score of -1.09, thus Kentucky has a 100pt score below 50.

Quartiles

For an added layer of nuance, quartiles are also available to compare states.

Quartile	Meaning	% of Scores
Q4	Best Scores	Top 25% : > 75th percentile
Q3	Above Average	Top 50%: 50 - 75th percentile
Q2	Below Average	Bottom 50%: 25th - 50th percentile
Q1	Worst Scores	Bottom 25%: 0 - 25th percentile

Scorecard Categories

For this process, scores were calculated for each of our three categories of focus in ADA-PARC: Community Living, Community Participation, Work Economic.

Community Living

Measures in this category are related to the living situation of the disabled population for each state.(-)Indicates measure is flipped since a higher value indicates worse outcomes for people with disabilities.

Indices in Community Living Scorecard

Name of Index	Variable 1	Variable 2
Communal Living Index	% of PWD Living at Home	% of PWD Living in Non-Institutionalized Group Quarters
Institutional Living Index	% of PWD in Living in Nursing Homes (-)	% of PWD Living in Correctional Facilities (-)

Measures in Community Living Scorecard

- Home: % of disabled population living at home.
- Nursing Home: % of disabled population age 18+ living in nursing home.
- Non-Institutionalized Group Quarters: % of disabled population living in non-institutionalized group quarters.
- Correctional Facility: % of disabled population in a correctional facility.

Community Participation

Measures in this category are related to means through which the disabled population is empowered to participate in society. (-)Indicates measure is flipped since a higher value indicates worse outcomes for people with disabilities.

Indices in Community Participation Scorecard

Name of Index	Variable 1	Variable 2	Variable 3
Tech Access Index	PWD % Access to Computer	PWD % Access to Smartphone	PWD % Access to Internet
Insurance Access Index	PWD % Public Insurance	PWD % Uninsured (-)	PWD % Private Insurance
Education Access Index	PWD % Less Than High School	PWD % Bachelors Degree	-
Transportation Index	PWD % Commute to work by Public Transit	PWD % Commute to Work by Car Alone (-)	-

Measures in Community Participation Scorecard

- Uninsured : % of disabled population age 19+ without health insurance.
- Public Insurance: % of disabled population age 19+ with public health insurance.
- Private Insurance: % of disabled population age 19+ with private health insurance.
- Commute Public Transportation: % of working disabled population commuting by public transportation.
- Commute Car Alone: % of working disabled population commuting by car alone.
- Less Than High School: % of disabled population age 25+ whose highest level of education is less than high school / GED equivalent.
- Bachelors Degree: % of disabled population age 25+ whose highest level of education is bachelors degree plus.
- Computer Access: % of disabled population with access to a computer.
- Internet Access: % of disabled population with internet access.
- Smartphone Access: % of disabled population with smartphone access.

Work Economic

Measures in this category are related to labor force characteristics of the disabled population. (-)Indicates measure is flipped since a higher value indicates worse outcomes for people with disabilities.

Indices in Work Economic Scorecard

Name of Index	Variable 1	Variable 2
Employment Index	PWD % Employed	PWD % Not in Labor Force (-)
Wealth & Poverty Index	PWD % Below Poverty (-)	PWD Median Income

Measures in Work Economic Scorecard

- Below Poverty: % of disabled population living below poverty line.
- Employment: % of working age disabled population that is employed.
- Unemployment % of working age disabled population that is unemployed.
- Median Income: median individual income for working age disabled population.
- Not In Labor Force: % of working age disabled population not in the labor force

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Contact Information

We would love to hear from you!

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